

Christopher Musco

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Academic Positions

New York University <i>Assistant Professor, Tandon School of Engineering</i> <i>Department of Computer Science and Engineering</i>	New York, NY <i>2019 - present</i>
Princeton University <i>Research Instructor, Department of Computer Science</i>	Princeton, NJ <i>2018 – 2019</i>
Massachusetts Institute of Technology <i>Research and Teaching Assistant, Theory of Computation Group</i>	Cambridge, MA <i>2013 – 2018</i>
Yahoo Labs <i>Research Intern, Scalable Machine Learning Group</i>	New York, NY <i>Summer 2015</i>

Research Areas

foundations of data science • scalable machine learning • numerical linear algebra • theory of algorithms
randomization • dimensionality reduction • sketching & streaming • spectral graph algorithms • optimization

Education

Massachusetts Institute of Technology <i>Ph.D. in Computer Science</i> Advisor: Jonathan Kelner Thesis: Faster Linear Algebra for Data Analysis and Machine Learning M.S. Thesis: Dimensionality Reduction for Sparse and Structured Matrices	Cambridge, MA <i>2013 – 2018</i>
Yale University <i>B.S. in Applied Mathematics, B.S. in Computer Science</i>	New Haven, CT <i>2008 – 2012</i>

Publications

*Authors appear in alphabetical order, in the tradition of mathematics and theoretical computer science. Any exceptions are marked with a *, which indicates that authors are ordered by contribution.*

Near Optimal Linear Algebra in the Online and Sliding Window Models. Vladimir Braverman, Petros Drineas, Cameron Musco, Christopher Musco, Jalaj Upadhyay, David P. Woodruff, Samson Zhou. *In submission.* 2020.

The Statistical Cost of Kernel Hyperparameter Tuning. Raphael A. Meyer, Christopher Musco. *In submission.* 2020.

Towards Equitable Transit Planning*. Sheng Wang, Yuan Sun, Christopher Musco, Zhifeng Bao. *In submission.* 2020.

Finding the Mode of a Kernel Density Estimate. Jasper C.H. Lee, Jerry Li, Christopher Musco, Jeff M. Phillips, Wai Ming Tai. *In submission.* 2020.

Low-Rank Toeplitz Matrix Estimation via Random Ultra-Sparse Rulers. Hannah Lawrence, Jerry Li, Cameron Musco, Christopher Musco. *International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2020.

Sample Efficient Toeplitz Covariance Estimation. Yonina Eldar, Jerry Li, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Fast and Space Efficient Spectral Sparsification in Dynamic Streams. Michael Kapralov, Aida Mousavifar, Cameron Musco, Christopher Musco, Navid Nouri, Aaron Sidford, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2020.

Understanding Filter Bubbles and Polarization in Social Networks. Uthsav Chitra, Christopher Musco. *ACM International Conference on Web Search and Data Mining (WSDM)* 2020.

Low-Rank Approximation from Communication Complexity. Cameron Musco, Christopher Musco, David Woodruff. *In submission*. 2019.

A Universal Sampling Method for Reconstructing Signals with Simple Fourier Transforms. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *ACM Symposium on Theory of Computing (STOC)* 2019.

Learning Networks from Random Walk-Based Node Similarities. Jeremy Hoskins, Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *Conference on Neural Information Processing Systems (NeurIPS)* 2018.

Eigenvector Computation and Community Detection in Asynchronous Gossip Models. Frederik Mallmann-Trenn, Cameron Musco, Christopher Musco. *International Colloquium on Automata, Languages, and Programming (ICALP)* 2018.

Minimizing Polarization and Disagreement in Social Networks. Cameron Musco, Christopher Musco, Charalampos Tsourakakis. *The Web Conference (WWW)* 2018.

Stability of the Lanczos Method for Matrix Function Approximation. Cameron Musco, Christopher Musco, Aaron Sidford. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2018.

Recursive Sampling for the Nyström Method. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2017.

Random Fourier Features for Kernel Ridge Regression: Approximation Bounds and Statistical Guarantees. Haim Avron, Michael Kapralov, Cameron Musco, Christopher Musco, Ameya Velingker, Amir Zandieh. *International Conference on Machine Learning (ICML)* 2017.

Input Sparsity Time Low-Rank Approximation via Ridge Leverage Score Sampling. Michael B. Cohen, Cameron Musco, Christopher Musco. *ACM-SIAM Symposium on Discrete Algorithms (SODA)* 2017.

Determining Tournament Payout Structures for Daily Fantasy Sports. Christopher Musco, Maxim Sviridenko, Justin Thaler. *SIAM Algorithm Engineering and Experiments (ALENEX)* 2017.

Principal Component Projection Without Principal Component Analysis. Roy Frostig, Cameron Musco, Christopher Musco, Aaron Sidford. *International Conference on Machine Learning (ICML)* 2016.

Randomized Block Krylov Methods for Stronger and Faster Approximate Singular Value Decomposition. Cameron Musco, Christopher Musco. *Conference on Neural Information Processing Systems (NeurIPS)* 2015. Invited for full oral presentation (1 of 15 out of 403 accepted papers). Also presented at the 2016 Copper Mountain Conference on Iterative Methods

Dimensionality Reduction for K-Means Clustering and Low Rank Approximation. Michael B. Cohen, Samuel Elder, Cameron Musco, Christopher Musco, Madalina Persu. *ACM Symposium on Theory of Computing (STOC)* 2015.

Principled Sampling for Anomaly Detection. Brendan Juba, Fan Long, Christopher Musco, Stelios Sidiropoulos-Douskos, Martin Rinard. *Network and Distributed System Security Symposium (NDSS)* 2015.

Uniform Sampling for Matrix Approximation. Michael B. Cohen, Yin Tat Lee, Cameron Musco, Christopher Musco, Richard Peng, Aaron Sidford. *Innovations in Theoretical Computer Science (ITCS)* 2015.

Single Pass Spectral Sparsification in Dynamic Streams. Michael Kapralov, Yin Tat Lee, Cameron Musco, Christopher Musco, Aaron Sidford. *IEEE Symposium on Foundations of Computer Science (FOCS)* 2014, Special Issue SIAM Journal on Computing 2017.

Student Research Advising

Raphael Meyer. Ph.D., New York University.	2019 - present
Prathamesh Dharangutte. M.S., New York University.	2019 - present
John Shin. M.S., New York University.	2019 - present
Tristan Pollner. Undergraduate, Princeton.	2018 - 2019

Teaching

Introduction to Machine Learning (NYU CS-UY 4563) <i>Course Instructor</i>	Spring 2020
Algorithmic Machine Learning and Data Science (NYU CS-GY 9223I) <i>Course Instructor</i>	Fall 2019
Advanced Algorithm Design (Princeton COS 521) <i>Course Instructor</i>	Fall 2018
Advanced Algorithms (MIT 6.854/18.415) <i>Teaching Assistant</i>	Spring 2016
Technical Communication Skills for Graduate Students (MIT 6.S977) <i>Workshop Leader</i>	Spring 2016
Introduction to Computer Science (Yale CPSC-201) <i>Teaching Assistant</i>	Spring 2012
Introduction to Programming (Yale CPSC-112) <i>Teaching Assistant</i>	Fall 2011

Mentorship and Outreach

SIAM/MathWorks Math Modeling Challenge
Problem Development Committee Member, Judge 2018 – present
Support SIAM's annual math modeling competition, which gives high school students an opportunity to apply mathematics, statistics, and data science to a real-world problem. Serve on committee in charge of writing practice and contest problems. Score and provide feedback on student submissions to the contest.

MIT Graduate Communication Lab
Communication Adviser 2015 – 2018
Helped to establish an EECS Communication Lab to provide over 700 graduate students with support in academic writing, oral presentation, job applications, and generally, in communicating effectively. Offered feedback, guidance, and coaching to individual students during weekly office hours.

CSAIL Algorithms Office Hours
Member 2016 – 2018
Advised researchers, generally in applied fields, on framing and solving algorithmic problems arising in their work.

Research Science Institute at MIT
Summer Research Adviser 2014

Primary supervisor for a high school student participating in the prestigious RSI summer research program. Advisee's project, "Relaxation of a Concurrent Disjoint-Set", was selected as one of 10 program finalists.

Talks and Presentations

Introduction to Machine Learning

New York University, guest lecture in "Introduction to Computer Science" *Mar. 2020*

Structured Covariance Estimation

Yahoo! Research, Seminar *April 2020*

Johns Hopkins University, Theory Seminar *Oct. 2019*

Symposium on Discrete Algorithms (SODA) *Jan. 2020*

From Randomized Linear Algebra to Randomized Functional Analysis

Rutgers University, DIMACS RandNLA Workshop *Sept. 2019*

University of Utah, Data Science Seminar *Jan. 2020*

Learning Signals with Simple Fourier transforms

Symposium on Theory of Computing (STOC) *June 2019*

Columbia University, Theory Seminar *Oct. 2018*

UC Berkeley, Simons Institute Workshop *Sept. 2018*

Princeton University, Theory Seminar *Sept. 2018*

The Lanczos Method in Data Science: Challenges and Opportunities

SIAM Conference on Parallel Processing for Scientific Computing, Minisymposium *Feb. 2020*

Georgetown University, CS Seminar *Feb. 2018*

Symposium on Discrete Algorithms (SODA) *Jan. 2018*

Unifying Random Features and Leverage Scores for Kernel Matrix Approximation

Institute for Advanced Study, Computer Science and Discrete Mathematics Seminar *Feb. 2018*

Harvard University, guest lecture in "Sketching Algorithms for Big Data" *Nov. 2017*

Microsoft Research, Redmond *Sept. 2017*

Sketching and Sampling Methods for Large Scale Linear Algebra

Twitter, Cambridge *July 2017*

Recursive Sampling for the Nyström Method

Neural Information Processing Systems (NeurIPS) poster session *Dec. 2017*

New England Machine Learning Day poster session *May 2017*

Algorithms for Determining Tournament Payout Structures

Meeting on Algorithm Engineering and Experiments (ALENEX) *Jan. 2017*

Introduction to Linear Sketching

MIT, Theory Retreat *Sept. 2016*

Principal Component Projection without Principal Component Analysis

International Conference on Machine Learning (ICML) *June 2016*

Ridge Leverage Score Sampling

National Institute of Informatics, Shonan Meeting *July 2016*

MIT, Algorithms & Complexity Seminar *July 2016*

University of Utah, Data Group Meeting *Jan. 2016*

Randomized Block Krylov Methods

Neural Information Processing Systems (NeurIPS) poster session

Dec. 2015

Large Scale Column Subset Selection

Yahoo Labs, Science Week poster session

July 2015

Dimensionality Reduction for k-Means Clustering

IBM T.J. Watson, Mathematical Sciences research seminar

Aug. 2015

Symposium on Theory of Computing (STOC)

June 2015

MIT, Theory Lunch

June 2014

Principled Sampling for Anomaly Detection

MIT, Defense Advanced Research Projects Agency site visit

Mar. 2015

Network and Distributed System Security Symposium (NDSS)

Feb. 2015

Uniform Sampling for Matrix Approximation

Yahoo Labs, Scalable Machine Learning research seminar

June 2015

Single Pass Spectral Sparsification in Dynamic Streams

MIT, Annual Sublinear Algorithms Day poster session

Apr. 2015

Harvard University, Theory Seminar

Nov. 2014

Foundations of Computer Science (FOCS)

Oct. 2014

Service

Conference Programming.....

NeurIPS Area Chair

2019

Organization.....

SIAM Mathematics of Data Science Minisymposium Organizer

2020

NYU Tandon Algorithmic Data Science Reading Group

2019 - present

Department Service.....

Faculty Search Committee

2019

Ph.D. Admissions Committee

2019 - present

Conference Reviewing.....

STOC, FOCS, SODA, ITCS, ICALP, ESA, ICML, NeurIPS, APPROX, RANDOM, STACS, IWOCA, WALCOM, NDSS

Journal Reviewing.....

Journal of Machine Learning Research, SIAM Journal on Matrix Analysis and Applications, SIAM Journal on Scientific Computing, SIAM Journal on Mathematics of Data Science, Science Advances, Mathematical Programming, IEEE Transactions on Signal Processing, Advances in Data Analysis and Classification, IEEE Transactions on Knowledge and Data Engineering, International Journal of Foundations of Computer Science.

Grants Panels.....

NSF Division of Information and Intelligent Systems (2019)

Notables.....

ICML 2018 Outstanding Reviewer

Other Research Experience

Rutgers, DIMACS

Invited participant

2019

Workshop on Randomized Numerical Linear Algebra, Statistics, and Optimization.

UC Berkeley, Simons Institute for the Theory of Computing

Invited participant

2018

Workshop on Randomized Numerical Linear Algebra and Applications.

Japan National Institute of Informatics

Invited participant

2016

Shonan meeting on Recent Advances in Randomized Numerical Linear Algebra.

Honors and Awards

National Science Foundation

2014-2017

Graduate Research Fellowship, recipient.

Yale University

2012

Cum Laude, Distinction in Both Majors.

Other Professional Experience

Redfin

Seattle, WA

Software Engineer, Data Team

2012 – 2014

Developed backend Java infrastructure for internet-powered real estate startup. Incorporated large-scale machine learning algorithms into production code. Managed three interns and onboarded four new-hires.

Elysium Digital

Cambridge, MA

Summer Consultant

2011

Provided technical expertise in software patent cases at leading litigation consulting company.

Amicus

New Haven, CT

Software Developer

2010 – 2011

Built applications for political campaign management and social fundraising at an early-stage startup that went on to raise 4 million dollars in seed funding.